ETF Flows and Stock Market Dynamics: Patterns, Signals and Portfolio Construction Insights

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Abstract

The rapid growth of exchange-traded funds (ETFs) has raised important questions about the role of ETF flows in the pricing and performance of underlying equities. Using the transparency of the ETF structure, we examine the relationship between past ETF flows and subsequent stock returns. While short-term mean reversion effects are well documented, evidence on longer-term effects remains limited.

Based on a broad dataset covering a substantial share of the global ETF market capitalization, we find that flows from thematic and sectoral ETFs are associated with distinct return patterns across investment horizons, with short-term reversal effects and longer-term trend-following dynamics particularly visible in U.S. equities.

Building on these findings, we construct a systematic cross-sectional equity signal that combines short- and long-horizon ETF flow information. The resulting portfolios exhibit meaningful return dispersion and low correlation with traditional equity factors, suggesting that ETF flows may provide a complementary source of information for stock selection and systematic portfolio construction. The effects are weaker in European equities, where ETF ownership remains lower.

Keywords– ETF, ETF flows, ETF ownership, factor investing, smart beta, cross-sectional stock returns, thematic investing, portfolio construction

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1 Introduction

As many segments of the market seem to become more flow-driven than fundamentally driven, an expanding strand of the academic literature has focused on the impact of fund flows on underlying equity markets. Whether in the context of mutual funds or ETFs, this issue has been a key concern for researchers for years.

Prior research has examined the impact of mutual fund flows on the returns of underlying securities. Warther (1995) is among the first to examine the relationship between mutual fund flows and subsequent security returns. He studies this link across equities, bonds, and gold, and finds that unusually large inflows are contemporaneously associated with higher security returns. However, he reports little evidence that expected flows have predictive power for future returns. Coval and Stafford (2007) also document that large capital outflows force mutual funds to liquidate commonly held positions, generating significant downward price pressure, so-called "fire sales", while large inflows similarly induce price pressure through coordinated purchases. Their evidence suggests that flow-induced trading can temporarily push stock prices away from fundamentals, even in highly liquid equity markets. Beyond short-term price pressure, the authors provide evidence of a longer-term trend-following dynamic. When inflow-induced purchases are broad relative to available sellers, coordinated institutional trading creates persistent price distortions that endure for several months, typically around two quarters, and dissipate only slowly over subsequent quarters.

With the growing importance of the ETF market, several authors have extended similar analyses specifically to ETFs. Global ETF assets under management reached an all-time high of $18 trillion in 2025[1], following a prolonged period of exponential growth. Over the past decade, investors have increasingly adopted the ETF vehicle, attracted by its transparency, tax efficiency, cost efficiency, and intraday trading. The ETF market has evolved substantially and now spans virtually all major asset classes. It enables investors to access far more than traditional market benchmarks, offering exposure to factor strategies, thematic investments, and quantitative approaches, among others. More recently, the expansion of ETF assets under management has been driven in part by the development of active ETFs, which blend active portfolio management with the liquidity, transparency, and operational efficiency of the ETF structure. This fast-growing segment has outpaced the broader ETF market and encouraged new entrants to launch products within the ETF ecosystem.

From an academic perspective, the effect of ETF flows on the underlying equity market is of particular interest. Owing to their high level of transparency, ETFs provide granular data that facilitate empirical investigation. Furthermore, in contrast to mutual fund flows which may be partially absorbed through cash holdings, ETF flows mechanically result in proportional investment and divestment across the entire underlying portfolio.

As highlighted by Ben-David, Franzoni and Moussawi (2017), ETFs can have two opposing effects on stock prices. The additional liquidity they provide can enhance price discovery in the underlying securities, making prices more informative. At the same time, ETF trading that is not driven by fundamentals may spill over to the underlying securities, and potentially leading to mispricing. The authors suggest that these two opposing effects can occur simultaneously.

Several studies provide evidence that ETF flows generate non-fundamental price effects on underlying securities. Da and Shive (2018) show that ETF arbitrageurs act as a transmission channel for non-fundamental shocks to the equity market. As a result, stocks included in ETFs display stronger comovement, driven by investor reactions to index-related information rather than firm-specific news. Prior research also documents that ETF-induced trading activity can create temporary price pressure at the underlying securities level, which is followed by price reversals. Using an ETF flow definition that differs from ours, Staer (2017) finds evidence of a price reversal effect that fades after a few days. Likewise, Sandberg (2018) develops a model of ETF-induced price impact and documents a modest and transitory effect on underlying securities. The author further demonstrates that incorporating an ETF flow sensitivity factor into portfolio construction improves risk-adjusted performance.

While the literature has extensively documented short-term, non-fundamental price effects of ETF flows, evidence on their role in reinforcing longer-term market trends remains limited. The Deutsche Bundesbank, in its October 2018 Monthly Report (Deutsche Bundesbank 2018), warned that ETFs could amplify market movements during periods of correction. The report highlights that procyclicality in ETF investing can amplify existing market trends, as passive portfolio rebalancing may trigger mechanical selling of underlying securities following price declines, thereby reinforcing market movements and potentially increasing systemic risk during periods of stress.

Van der Beck, Bouchaud and Villamaina (2024) document that investors struggle to distinguish between flow-driven and fundamental returns, thereby reinforcing self-inflated returns through performance-chasing behaviour. The authors introduce the concept of *Ponzi funds*, showing that some investors are effectively chasing their own price impact. Studies pointing to increased comovement are also consistent with a trend-following mechanism.

This paper makes several contributions to the literature. First, it documents a short-horizon reversal pattern associated with ETF flows. Second, it shows that the predictive content of ETF flows differs materially across ETF types: longer-horizon positive return predictability is concentrated in flows from thematic and sectoral ETFs, while it is not robustly observed for the broader ETF universe. Third, it documents systematic behavioural patterns among ETF investors. Finally, it demonstrates that the information embedded in ETF flows can be exploited to construct economically meaningful signals for equity portfolio managers.

Most existing studies rely on substantially less comprehensive datasets and focus primarily on ETFs listed in the United States. To our knowledge, this study is the first to use a dataset of this breadth and to analyse ETF flows across a global universe of ETFs. While most studies also omit data on delisted ETFs, we were able to incorporate them into our analysis, thereby limiting survivorship bias.

2 Materials and Methods

2.1 Data

2.1.1 Data source

This study relies on ETF data provided by Trackinsight. As of 02/01/2026, the database covers 18,803 ETFs worldwide, of which 13,830 are currently open, while the remainder have been delisted. Delisted ETFs are kept in

[1] According to Trackinsight

the sample in order to mitigate survivorship bias and enable point-in-time analysis. The database comprises three categories of data :

1. Reference data, including 40 technical attributes (e.g., dividend policy, replication method, benchmark index) and 31 fields describing ETF strategies (e.g., asset class, thematic focus, sector exposure).
2. Daily time series, covering Net Asset Value (NAV), shares outstanding, dividends, splits, and flows.
3. Detailed holdings information for physically replicated ETFs, updated on a monthly basis.

Synthetic ETFs are excluded from the analysis because their flows do not necessarily translate directly into trading in the underlying stocks. The analysis period starts on 01/06/2020, corresponding to the earliest availability of holding data in our database.

Active exchange-traded funds (ETFs) are included in our sample, as they represent 31% of the ETFs in our database and account for approximately 10% of total assets under management. Their inclusion allows us to capture a broader measure of ETF-related ownership and, consequently, to better assess the aggregate effect of ETF activity on individual securities. It is important, however, to acknowledge the structural differences between active and passive ETFs. In particular, portfolio construction in active ETFs may lead to different portfolio adjustment dynamics in response to inflows and outflows.

2.1.2 ETF Ownership: Data and Stylized Facts

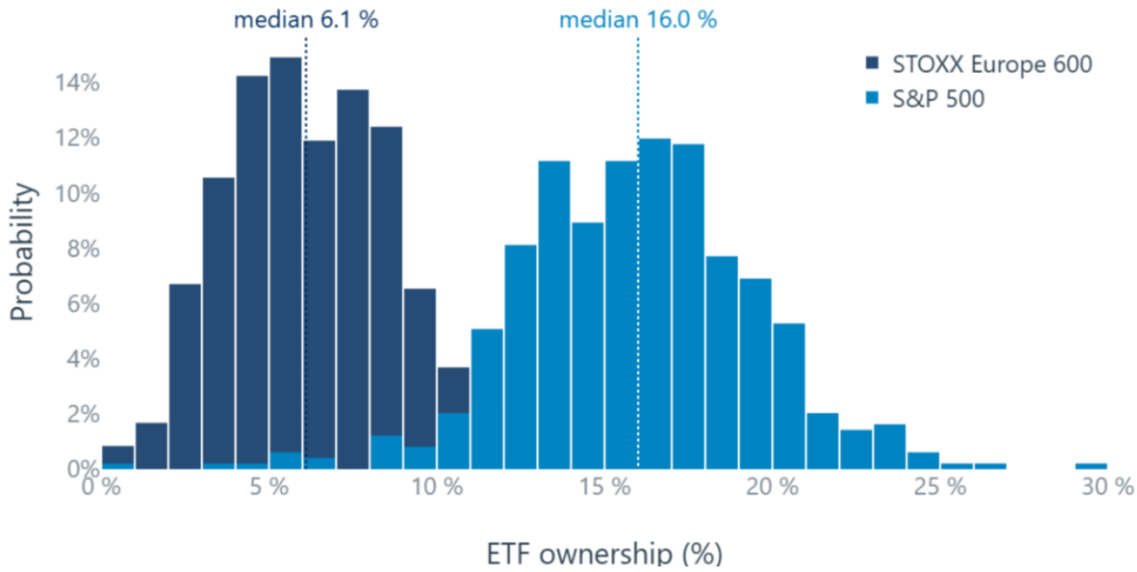
Our analysis focuses on S&P 500 constituents, with index membership updated monthly to reflect delistings and merger-and-acquisition events occurring between quarterly rebalancings. We focus on U.S. equities, as ETF ownership is significantly more important in the United States than in Europe. We define ETF ownership as the proportion of a company's market capitalization held by ETFs :

$$\text{ETF ownership}_i = \frac{\sum_{j=1} w_{i,j} * aum_j}{mc_i} \tag{1}$$

where $w_{i,j}$ represents the weight of the company i in the ETF j, aum_j represents the asset under management of ETF j and mc_i represents the market capitalization of the company i.

Figure 1 shows the distribution of ETF ownership among S&P 500 and STOXX 600 constituents. ETF ownership is significantly more prevalent in U.S. equities than in European equities, with median ownership of 16% for the S&P 500 versus 6.1% for the STOXX 600.

Figure 1: Distribution of ETF Ownership among S&P 500 and STOXX 600 companies



Source: Trackinsight, Bloomberg & authors' calculations.

Hence, the disparity between Europe and the United States is substantial. ASML, one of the European companies with the highest level of ETF ownership, has only 8.5% of its market capitalization held by ETFs, well below the median level observed among S&P 500 constituents.

An analysis of the time series dynamics of ETF ownership on the S&P 500 reveals a persistent increase in ETF stakes over time, as documented in Table 1. This trend underscores the growing influence of ETFs on the equity market and suggests that ETF ownership is becoming an increasingly significant factor in the valuation and liquidity of S&P 500 stocks.

These trends provide a natural motivation for examining whether ETF flows have a measurable impact on the return dynamics of individual stocks. These findings suggest that if signals derived from ETF activity can be used to identify US equity selection opportunities today, their relevance may strengthen over time as ETF ownership continues to expand.

Given that the United States can be viewed as a leading laboratory for ETF innovation and adoption, it is reasonable to expect that ETFs will exert an increasingly significant influence on stock returns across other geographies in the near future. As ETF penetration deepens in Europe, similar dynamics may emerge, potentially allowing for the application of comparable analytical frameworks to European equities.

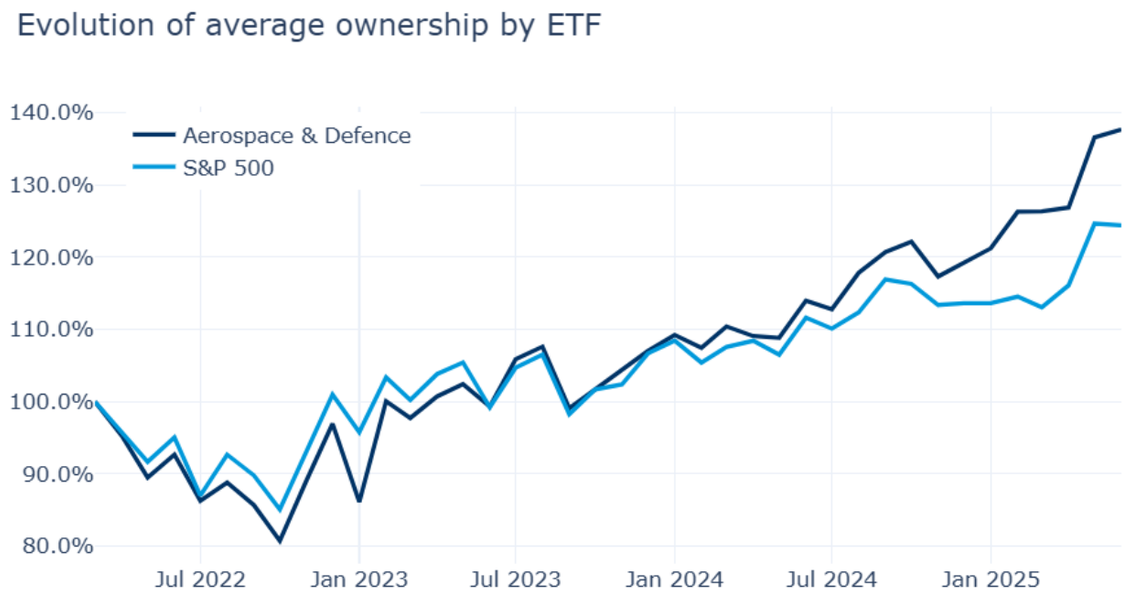
Another interesting stylised fact emerging from our analysis is that ETF ownership of stocks associated with popular thematic segments tends to increase faster than for the broader market. Figure 2 shows the evolution of

Table 1: Distribution of ETF Ownership among S&P 500 Constituents (%)

Year	First quartile	Median	Third Quartile
2020	9.9	11.5	13.5
2021	10.2	12.0	14.3
2022	10.9	12.9	15.3
2023	11.7	13.8	15.7
2024	12.3	14.7	16.8
2025	13.6	16.0	18.1

ETF ownership in Aerospace & Defence stocks relative to the rest of the market. As this theme gained popularity among investors, ETFs accumulated positions in Aerospace & Defence companies at a markedly faster pace than in other sectors. This pattern suggests that emerging investment trends become visible relatively quickly in ETF data.

Figure 2: Indexed Evolution of Average ETF Ownership for Defence Companies and the S&P 500



Source: Trackinsight, Bloomberg & authors' calculations.

2.2 Methodology

2.2.1 Classification of ETFs

Following Ben-David, Franzoni, Kim and Moussawi (2022), we classify ETFs according to their investment objectives, and distinguish four categories :

1. Thematic ETFs, as defined by the Trackinsight classification,
2. Sectoral ETFs, which track one or multiple sectors,
3. Smart beta ETFs, as identified by Trackinsight,
4. Other ETFs.

We adopt this categorisation to differentiate between market-access ETFs and specialised ETFs. The rationale is that flows into specialised ETFs may carry stronger informational content, as they more directly capture investors' underlying preferences. It is also worth noting that there is no overlap between sectoral and thematic ETFs in our methodology. We adopt this classification based on the intuition that strong trend-following behaviour can emerge in ETF markets, as observed during episodes such as the technology rally in 2024 or the defence sector boom in 2025. To assess this effect, we compute a correlogram based on Spearman rank correlations, which clearly highlights the presence of trend-following dynamics. Specifically, we examine the relationship between ETF flows into individual stocks over horizons ranging from 1 to 12 months and their subsequent monthly returns.

Depending on the specification, we consider either the full universe of ETFs or restrict to thematic and sectoral ETFs. We define ETF-induced flows for a given stock as :

$$ETFPressure_{i,t} = \frac{\sum_{j \in J} w_{i,j,t-m} * flow_{j,\,[t-m,\,t-1]}}{mc_{i,t-1}} \tag{2}$$

where t denotes the beginning of a calendar month, J denotes the set of ETFs considered, $w_{i,j,t-m}$ is the weight of company i in ETF j at the start of the month preceding t, $flow_{j,\,[t-m,\,t-1]}$ represents cumulative net flows into ETF j over that month, and $mc_{i,t-1}$ denotes the last market capitalization of the company i before date t.

This metric measures, for each stock at a given point in time, the cumulative pressure exerted over the past month through ETF flows. It reflects the extent to which a stock is exposed to buying or selling pressure associated with capital movements into and out of the ETFs that hold it.

Flows are computed from ETF share creations and redemptions in the market. We focus exclusively on primary market flows and disregard secondary market transactions, making our measure conceptually comparable to mutual fund flows, although structurally different. Since each secondary market trade involves a buyer and a seller for the

same quantity, these transactions do not represent directional bets at the aggregate level. Accordingly, we focus exclusively on net primary market inflows and outflows:

$$flow_{j,t} = NAV_{j,t} * (NOS_{j,t} - NOS_{j,t-1}) \tag{3}$$

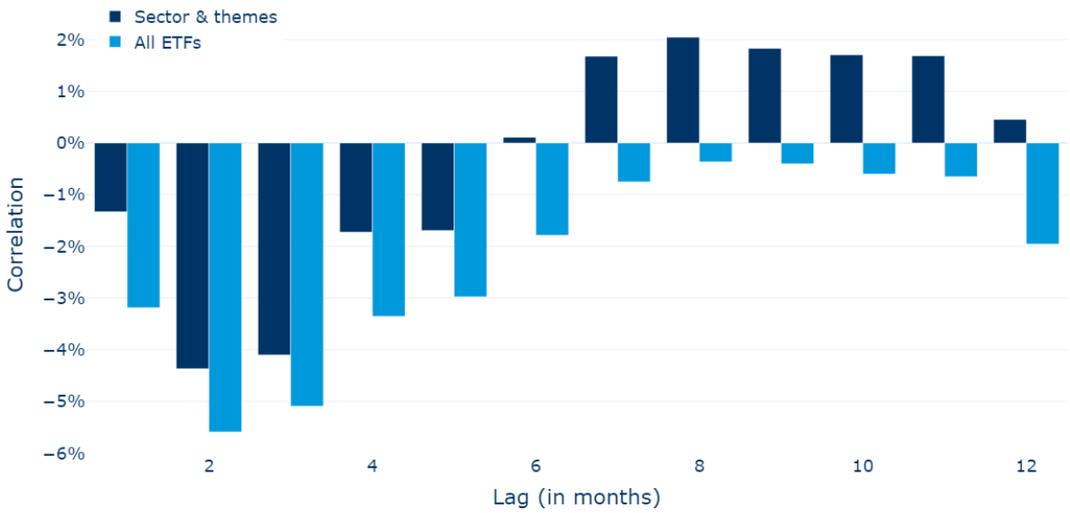
where t denotes a given date, $flow_{j,t}$ captures net creations or redemptions in ETF j, computed as the change in shares outstanding scaled by its net asset value, with $NAV_{j,t}$ denoting the ETF's net asset value and $NOS_{j,t}$ its number of shares outstanding.

Daily flows are aggregated to the monthly level. The following measure then captures the total net inflow or outflow into ETF j over the month preceding the date t:

$$flow_{j,\,[t-m,t-1]} = \sum_{\delta t=t-m}^{t-1} flow_{j,\delta t} \tag{4}$$

To assess whether each ETF category defined in our classification framework exerts a similar impact on the equity market, we conduct two sets of analyses: one based on the full dataset and another restricted to thematic and sectoral ETFs.

Figure 3: Spearman correlations of past cumulative ETF-induced flows and monthly returns



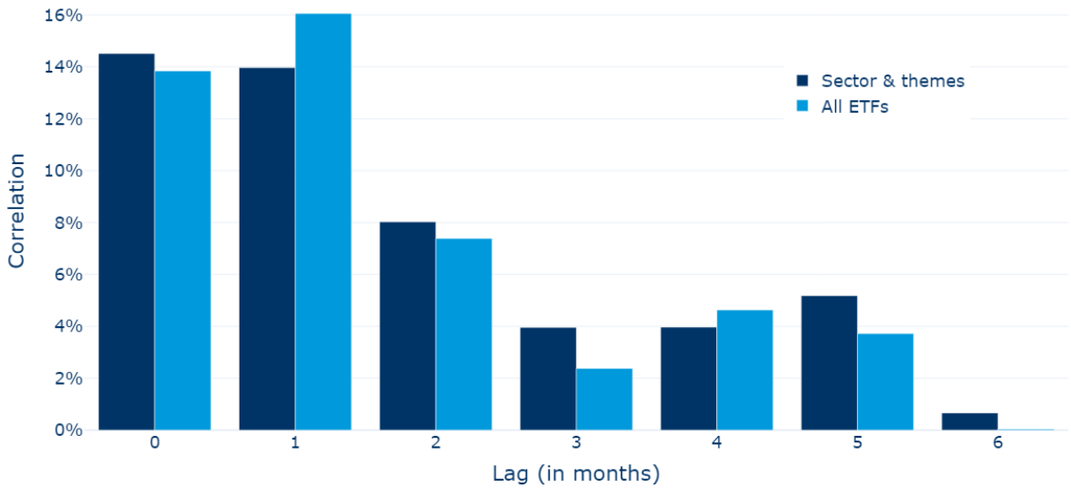
Source: Trackinsight, Bloomberg & authors' calculations.

Although the correlations are relatively small, the correlogram provides several informative insights. First, we observe a negative relationship between past ETF flows and subsequent returns over short horizons (1 to 5-6 months). This pattern is well documented in the academic literature. We observe this pattern both when flows are aggregated across the full dataset and only within the subsample of thematic and sectoral ETFs. At longer horizons (6–12 months), a trend-following effect appears, but only for flows from thematic and sectoral ETFs. Smart beta ETFs do not exhibit any distinct behaviour in our analysis, and we therefore exclude this category as a stand-alone group and merge it with the *Other ETF* category from the remainder of the analysis.

The correlogram analysis then suggests that flows into sectoral and thematic ETFs are more informative than flows into non-specialised ETFs. While this analysis provides useful intuition, these effects are formally tested and confirmed in the empirical results section.

Another behavioural pattern highlighted by our empirical analysis in Figure 4 is that ETF investors tend to allocate capital to recently well-performing products, indicating a pronounced trend-following behaviour.

Figure 4: Spearman correlations of returns and subsequent ETF-induced flows



Source: Trackinsight, Bloomberg & authors' calculations.

This effect has already been documented in the literature by Clifford, Fulkerson and Jordan (2014), who provide evidence of return-chasing behaviour among ETF investors. Coval and Stafford (2007) demonstrate that past performance is a significant driver of subsequent capital flows in the mutual fund industry. From this perspective, we do not detect any meaningful behavioural difference between mutual fund and ETF investors, as flows in both segments respond similarly to past returns.

2.2.2 Model

We formally examine the relationship suggested by the correlograms between past cumulative ETF pressure and subsequent stock returns, with the aim of providing rigorous statistical evidence to validate the observed patterns and assess their relevance for empirical research and practical applications.

The correlation analysis indicates that flow variations associated with thematic and sectoral ETFs contain substantially more informational content, whereas flows from other ETF categories exhibit weaker and less systematic relationships with returns. Accordingly, the subsequent empirical analysis concentrates on these two specifications, as the regression framework incorporates pressure features constructed either from the full universe of ETFs or exclusively from the thematic and sectoral ones, allowing for a direct comparison of their relative explanatory power.

Rather than including all accumulation horizons simultaneously in a single regression, we estimate a separate regression for each horizon. This approach avoids introducing highly correlated explanatory variables arising from overlapping flow-accumulation windows and allows the effect associated with each horizon to be assessed independently. Consistent with the correlogram analysis, this approach allows us to study the relationship between returns and cumulative flows at each horizon individually, with the objective of isolating and quantifying the statistical contribution of each phenomenon separately, improving overall interpretability.

Let $r_{i,[t,\,t+m]}$ denote the return of stock i during the month following date t, and let $ETFPressure_{\,i,\,[t-n\cdot m,\,t]}$ denote the cumulative ETF pressure on stock i during the n months preceding the date t :

$$ETFPressure_{i,\,[t-n\cdot m,\,t]} = \sum_{k=0}^{n-1} ETFPressure_{i,t-k\cdot m} \tag{5}$$

We also control for aggregate market conditions. Let $r^{Mkt}_{[t,\,t+m]}$ denote the return of the market during the month following date t, measured by the return on the S&P 500 in our empirical specification.

For each time lag n, we estimate the following regression across stocks and time:

$$r_{i,[t,\,t+m]} \;=\; \alpha \;+\; \beta \cdot ETFPressure_{\,i,\,[t-n\cdot m,\,t]} \;+\; \gamma \cdot r^{Mkt}_{[t,\,t+m]} \;+\; \epsilon_{i,t} \tag{6}$$

where i indexes stocks, t indexes time, n denotes the length of the accumulation window in months, α is a constant, β measures the sensitivity of stock returns to cumulative ETF-induced pressure, γ controls for market-wide movements, and $\varepsilon_{i,t}$ is an idiosyncratic error term.

Standard errors are computed under asymptotic normality assumptions, and all statistical tests are two-sided.

We account for aggregate market movements by including a market return control in the regression. Rather than modelling raw stock returns, we focus on returns net of broad market fluctuations. The coefficient γ captures the contribution of market-wide movements, so that the estimation of β relies on variation in stock returns beyond the market component. Empirically, omitting this control leads to lower statistical precision, indicating that accounting for market dynamics materially reduces noise. Overall, this specification allows us to estimate the cross-sectional effect associated with ETF-induced pressure across stocks.

3 Empirical results

3.1 Statistical analysis

To validate the intuition outlined in section 2.2.2, we estimate the model specified in Equation 6 using the Trackinsight database. We first conduct the analysis on the full sample of ETFs and subsequently restrict the universe

to sectoral and thematic ETFs to examine whether the long-term momentum effect is particularly pronounced within this subset from a statistical perspective.

We begin by conducting the regression analysis on the full ETF universe. Table 2 reports the estimation results of the regressions specified in Equation 6, where ETF pressures are aggregated over horizons ranging from one to twelve months.

We estimate Equation 6 separately for each accumulation window n ranging from one to twelve months, to avoid including highly correlated ETF-pressure measures constructed over overlapping horizons in the same regression. To account for aggregate market movements, we include a market return control so that identification primarily relies on cross-sectional variation in stock returns beyond the market component. Confidence intervals are reported at the 90% level and are consistent with the statistical significance indicated by the p-values. We adopt a 10% significance level in our regressions, which is standard practice in empirical finance studies of this type.

Table 2: Regression Results across Horizons (S&P 500, All ETFs)

Horizon	Coefficient	Standard Error	p-value	90% Confidence Interval	R^2	Observations
1	-1.03**	0.46	0.03	[-1.79, -0.27]	0.22	23,097
2	-0.56**	0.28	0.04	[-1.02, -0.10]	0.22	23,094
3	-0.36*	0.21	0.08	[-0.70, -0.02]	0.22	23,091
4	-0.18	0.17	0.28	[-0.45, 0.09]	0.22	23,087
5	-0.28**	0.14	0.05	[-0.51, -0.05]	0.22	23,083
6	-0.27**	0.12	0.03	[-0.47, -0.07]	0.22	23,078
7	-0.19*	0.11	0.08	[-0.36, -0.01]	0.22	23,072
8	-0.01	0.10	0.96	[-0.17, 0.16]	0.22	23,067
9	0.00	0.09	0.96	[-0.15, 0.14]	0.22	23,062
10	0.09	0.08	0.27	[-0.05, 0.23]	0.22	23,056
11	0.17**	0.08	0.04	[0.04, 0.30]	0.22	23,037
12	0.02	0.08	0.82	[-0.11, 0.14]	0.22	22,564

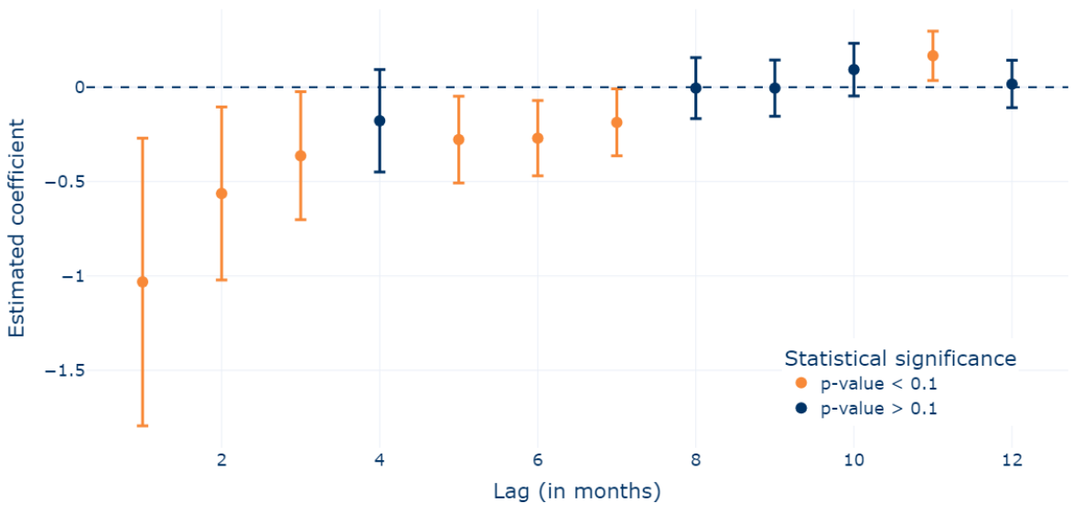
Notes: *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. All tests are two-sided.

The results reveal a clear asymmetry between short and long horizon effects. At short horizons, estimated coefficients are consistently negative, economically meaningful, and statistically significant at the 5% and 10% levels. Confidence intervals lie predominantly below zero, providing robust evidence of a short-term reversal effect. This pattern is strongest over horizons ranging from one to six months and remains detectable, though weaker, up to approximately eight months.

In contrast, there is little evidence of a statistically robust long-term effect. At longer horizons, estimated coefficients are small in magnitude and confidence intervals mostly span both sides of zero, indicating a lack of reliable directional relationship. Although a positive coefficient appears at the eleven-month horizon, this isolated result is not sufficient to support the existence of a persistent long-term effect. We do not observe any contiguous range of horizons over which coefficients are both positive and statistically significant.

The explanatory power of the regressions remains stable across specifications, with R^2 values around 22%. This level is consistent with the proportion of return variation typically captured by aggregate market movements. The objective of this analysis is not to maximise predictive performance, but rather to detect and assess the statistical significance of flow-driven effects. In that respect, the results are internally consistent: the short-term effect is clearly identified, whereas longer-term dynamics remain weak and statistically indistinguishable from noise.

Figure 5: Flow Coefficients and 90% Confidence Intervals by Horizon (S&P 500, All ETFs)



Source: Trackinsight, Bloomberg & authors' calculations.

The figure provides a visual confirmation of these results. Short-horizon coefficients are clearly negative and statistically significant, while longer-horizon estimates cluster around zero with wide confidence intervals. Overall, the evidence supports the presence of a short-term price pressure effect, with no robust indication of a persistent long-term relationship.

We next restrict the construction of ETF-induced pressure to thematic and sector-specific ETFs, in order to better capture flows reflecting investors' targeted allocation decisions. We estimate the same regression specification as in the baseline case across all accumulation windows.

Table 3: Regression Results across Horizons (S&P 500, Thematic and Sectoral ETFs)

Horizon	Coefficient	Standard Error	p-value	90% Confidence Interval	R^2	Observations
1	-0.83	0.78	0.29	[-2.10, 0.45]	0.22	22,212
2	-1.53***	0.50	< 0.01	[-2.35, -0.71]	0.22	22,210
3	-1.10***	0.38	< 0.01	[-1.71, -0.48]	0.22	22,208
4	-0.44	0.30	0.14	[-0.93, 0.05]	0.22	22,205
5	-0.66***	0.25	< 0.01	[-1.08, -0.24]	0.22	22,202
6	-0.29	0.22	0.19	[-0.65, 0.08]	0.22	22,198
7	0.04	0.20	0.84	[-0.29, 0.36]	0.22	22,193
8	0.29	0.18	0.11	[-0.01, 0.58]	0.22	22,189
9	0.29*	0.16	0.08	[0.01, 0.56]	0.22	22,185
10	0.41***	0.15	< 0.01	[0.16, 0.66]	0.22	22,180
11	0.49***	0.14	< 0.01	[0.26, 0.72]	0.22	22,158
12	0.31**	0.14	0.03	[0.08, 0.53]	0.22	21,702

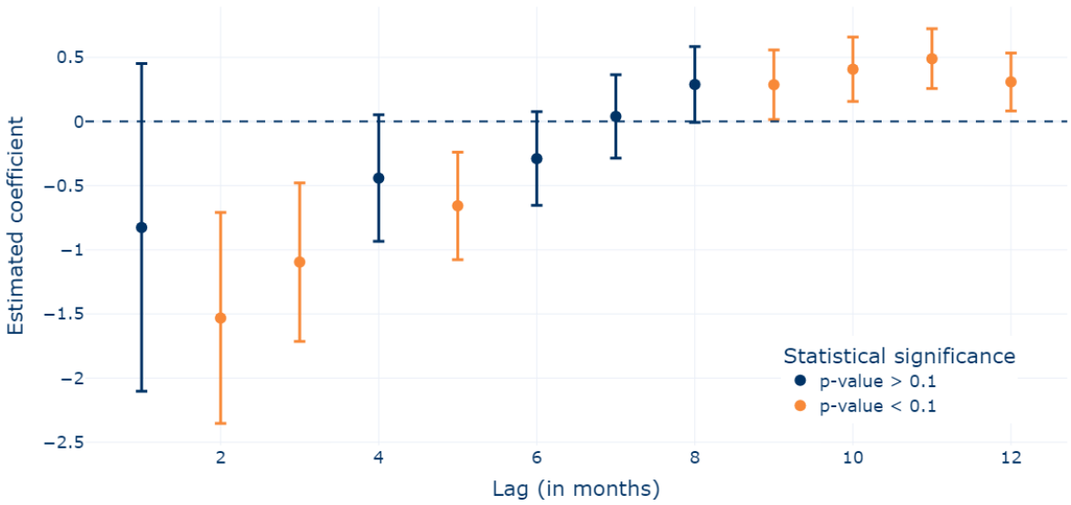
Notes: *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. All tests are two-sided.

The results differ markedly from the baseline specification and reveal a clear and structured pattern across horizons. At short horizons, especially at two and three months, ETF-induced pressure is associated with negative and statistically significant coefficients at the 1% level. Confidence intervals lie entirely below zero, providing strong evidence of a short-term reversal effect. This suggests that stocks experiencing strong inflows from thematic and sectoral ETFs tend to underperform in the subsequent period, consistent with temporary price pressure.

In contrast, a distinct long-term effect emerges at longer horizons. From nine to twelve months, estimated coefficients become positive, economically meaningful, and statistically significant at conventional levels. For example, the coefficient associated with ETF-induced pressure accumulated over a one-year horizon is statistically significant with a p-value of 0.03. Confidence intervals lie entirely above zero over this range, indicating a robust positive association between past ETF-induced pressure and subsequent returns. This pattern points to a longer-term effect in which flows into thematic and sectoral ETFs are associated with stocks that subsequently exhibit stronger performance.

Figure 6 then reports the estimated flow coefficients across all aggregation windows together with their 90% confidence intervals and indicates their statistical significance.

Figure 6: Flow Coefficients and 90% Confidence Intervals by Horizon (S&P 500, Thematic & Sectoral ETFs)



Source: Trackinsight, Bloomberg & authors' calculations.

The figure provides a clear visual confirmation of this dual structure. The separation between these regimes highlights the absence of a uniform effect across horizons and reinforces the presence of two distinct mechanisms.

Overall, ETF-induced flows from thematic and sectoral ETFs exhibit two distinct horizon-dependent effects. At short horizons, the results are consistent with temporary flow-induced price pressure that subsequently reverses. At longer horizons, they are positively associated with subsequent returns, suggesting that they reflect more persistent

allocation trends toward stocks that ultimately outperform. These findings indicate that ETF flows should be interpreted as a conditional cross-sectional signal whose economic impact depends critically on the investment horizon.

This distinction provides a natural foundation for strategy design, where short-term effects may be treated as transitory distortions, while longer-horizon flow information can be exploited as a signal of sustained investor demand.

3.2 Temporal Structure and Oscillatory Patterns in Flow Effects

Correlograms and horizon-by-horizon regressions indicate that the relationship between ETF-induced flows and subsequent stock market returns is neither monotonic nor independent from one aggregation window to another, but rather oscillates between opposing phenomena. This dual structure is most clearly visible when ETF-induced pressure is constructed using thematic and sector-specific ETFs, and we therefore use the same pressure measure in the sinusoidal specification below.

To test formally for such an oscillatory structure, we model the horizon-dependent effect as a low-frequency sinusoidal function of the time lag n, and we weight lagged ETF pressures accordingly. The resulting empirical profile then resembles an oscillatory adjustment rather than a linear propagation of effects.

We incorporate this structure directly into the following regression by replacing the cumulative flow term in Equation 6 with a sinusoidally weighted sum of past ETF pressures:

$$r_{i,[t,\,t+m]} = \alpha + \kappa \sum_{n=1}^{12} \cos(2\pi f n + \phi)\, ETFPressure_{i,\,t-n\cdot m} + \gamma \cdot r^{Mkt}_{[t,\,t+m]} + \epsilon_{i,t}, \tag{7}$$

where i indexes stocks and t indexes dates, α is a constant, κ scales the amplitude of the oscillatory component, f is the frequency and ϕ is the phase shift, and $\varepsilon_{i,t}$ is an idiosyncratic error term.

Rewriting the sinusoid using $\kappa\cos(x+\phi) = \beta_{\sin}\sin(x) + \beta_{\cos}\cos(x)$, the specification can be rewritten in a form that is linear in parameters and therefore directly estimable by linear regression:

$$r_{i,[t,\,t+m]} = \alpha + \beta_{\sin}\cdot\left(\sum_{n=1}^{12}\sin(2\pi f n)\, ETFPressure_{i,\,t-n\cdot m}\right) + \beta_{\cos}\cdot\left(\sum_{n=1}^{12}\cos(2\pi f n)\, ETFPressure_{i,\,t-n\cdot m}\right) + \gamma\cdot r^{Mkt}_{[t,\,t+m]} + \epsilon_{i,t}. \tag{8}$$

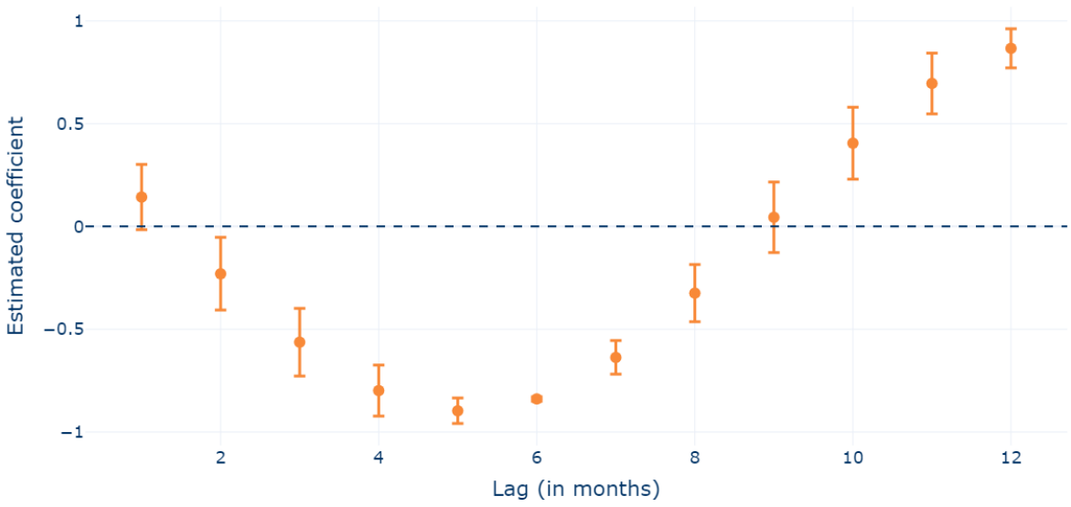
where i indexes stocks and t indexes dates, α is a constant, β_{sin} and β_{cos} scale the amplitudes of the oscillatory components, f is the frequency, and $\varepsilon_{i,t}$ is an idiosyncratic error term.

The frequency f is selected via a grid search, and the amplitude and phase of the sinusoid are recovered from the estimated coefficients $(\beta_{\sin}, \beta_{\cos})$.

This specification evaluates the statistical link between returns and a low-frequency Fourier representation of ETF-induced pressures. The objective is to determine whether the horizon-dependent structure observed in the baseline regressions can be parsimoniously summarized by a single oscillatory component. From an investment perspective, this parsimonious representation is particularly useful, as it allows the temporal structure of flow effects to be summarized by a small number of interpretable parameters, facilitating implementation in systematic allocation frameworks.

Figure 7 presents the cumulative sinusoidally weighted flow coefficients implied by the regression, together with their 90% confidence intervals.

Figure 7: Sinusoidal Fit of Flow Effects (S&P 500, Thematic and Sectoral ETFs)



Source: Trackinsight, Bloomberg & authors' calculations.

The figure confirms that the relationship between ETF-induced pressure and subsequent returns follows a structured oscillatory pattern across horizons. The key point is that the transition from a fully negative regime to a fully positive one is consistent with the estimated oscillatory period of 16 months, i.e. 8 months between two extremes.

The sinusoidal regression provides direct statistical support for this interpretation. Both β_{sin} and β_{cos} are significant at the 1% level, indicating that the oscillatory component is itself statistically significant. In other words, the cyclical structure observed in the horizon-by-horizon regressions is not a visual artefact, but a measurable feature of the data.

The specification also improves explanatory power. The adjusted R^2 rises, to approximately 23%, indicating that returns are better explained when ETF-induced pressure is weighted according to this estimated temporal structure than when pressures are accumulated uniformly across months. This result is economically intuitive. If flow effects alternate across horizons, imposing equal weights across lags necessarily masks part of the signal.

Overall, the sinusoidal specification shows that ETF-induced pressure does not affect returns in a uniform way over time. Its effect evolves in a systematic cycle, consistent with the coexistence of short-term price pressure and longer-horizon allocation effects.

3.3 An Application to Equity Portfolio Management

3.3.1 Simulation of historical performances

Our statistical analysis indicates that ETF flows contain measurable information about subsequent stock returns within the S&P 500 universe. This finding has material implications for equity portfolio managers operating in this market. Accordingly, we conduct a backtest of our proposed signal to evaluate its economic relevance and practical usefulness from a portfolio management perspective.

We combine the two effects identified in the statistical analysis to exploit both patterns observed in the data. The long portfolio targets stocks that have experienced substantial ETF inflows over the past year but limited inflows over the most recent three months, while the short portfolio focuses on the opposite configuration.

In accordance with the results of the statistical analysis, the signal is constructed exclusively from flows into thematic and sectoral ETFs. We implement a long–short equity strategy using a portfolio construction approach similar to that employed in the development of Fama–French factors. To mitigate potential backtest overfitting, we retain a twelve-month window to capture the long-term pressure effect component and a three-month window to capture the short-term component, although alternative horizons (11 and 2 months) produce slightly stronger statistical results.

ETF flows measured over three- and twelve-month horizons are scaled by market capitalization and converted into cross-sectional z-scores. The signal is defined as the difference between the twelve-month and three-month standardized pressure measures:

$$S_t = Z^{CS}\big(ETFPressure_{\,\cdot,\,[t-12\cdot m,\,t]}\big) - Z^{CS}\big(ETFPressure_{\,\cdot,\,[t-3\cdot m,\,t]}\big) \tag{9}$$

where for any cross-sectional vector $X \in \mathbb{R}^n$,

$$Z^{CS}(X)_i = \frac{X_i - \bar{X}}{\sigma(X)}$$

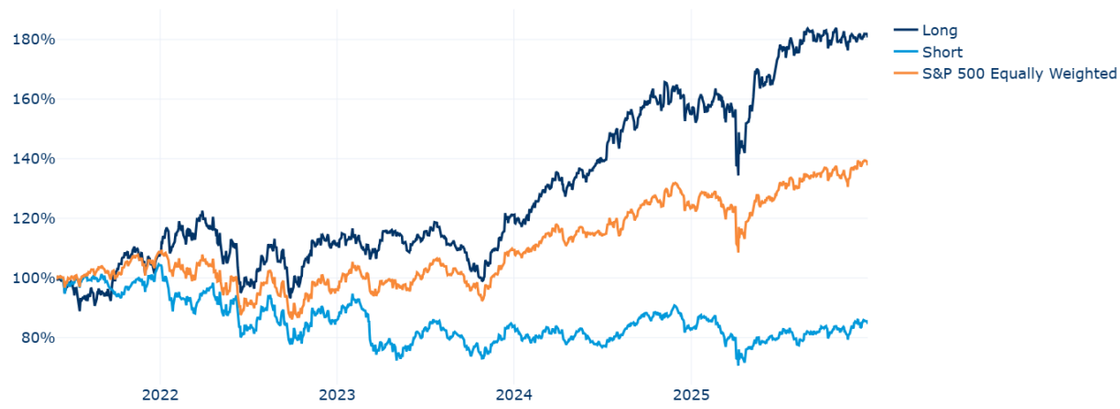
where i indexes stocks, and $\bar{X}$ and $\sigma(X)$ denote the cross-sectional mean and standard deviation of X.

Stocks in the S&P 500 are ranked according to this signal. The long (short) portfolio comprises the 50 stocks with the highest (lowest) signal values. Portfolios are rebalanced monthly and equally weighted.

The backtests presented here are purely hypothetical. In certain market conditions, implementing the short leg of the strategy may not be feasible in practice. The purpose of this exercise is therefore not to propose an immediately implementable trading strategy, but rather to demonstrate that the signal is informative and potentially valuable for equity stock selection.

Figure 8 indicates that the signal is informative on both the long leg and short leg. The long leg outperforms the S&P 500 Equally Weighted, while the short leg exhibits pronounced underperformance, clearly identifying stocks to avoid for users of the signal.

Figure 8: Performance of the Long and Short Portfolios based on the ETF Flow Signal



Source: Trackinsight, Bloomberg & authors' calculations.

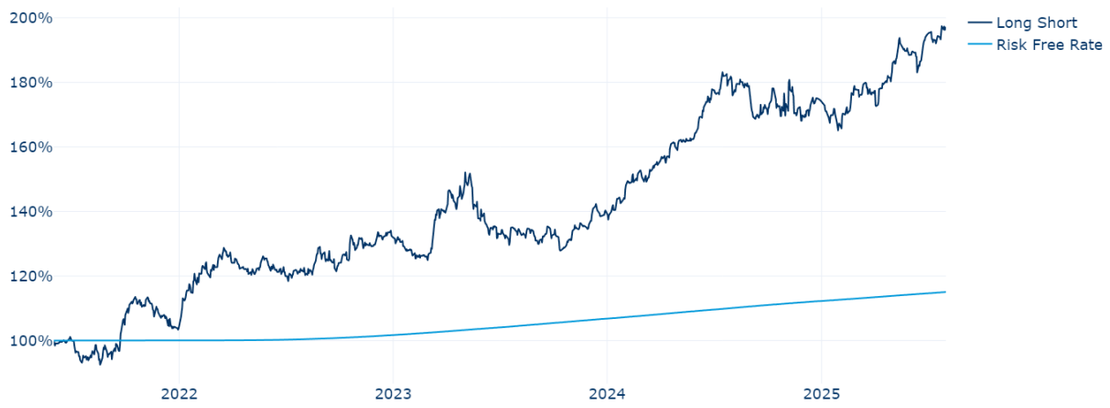
Table 4 shows that the long leg outperforms the benchmark by 7.6% per year, while the short leg underperforms by 10.3% per year. The long leg exhibits a higher Sharpe ratio than the S&P 500 Equally Weighted, while the short leg performs substantially worse on a risk-adjusted basis.

Table 4: Performance Statistics of the Long and Short Portfolio Legs

	Long Leg	Short Leg	S&P 500 Equally Weighted
Annualized Return (%)	13.44%	-4.45%	5.82%
Annualized Volatility (%)	20.63%	19.47%	17.35%
Maximum Drawdown (%)	-23.56%	-30.58%	-21.59%
Sharpe Ratio	0.49	-0.40	0.14

Examining the performance of the long–short portfolio in isolation is particularly informative and suggests the presence of a distinct risk premium. The volatility of the long–short portfolio is relatively high, reflecting the instability of its market beta. This outcome is not surprising given the construction of the strategy. The long leg tends to exhibit a low beta when defensive sectors, such as Utilities or Healthcare, experience substantial inflows. Conversely, it tends to display a higher beta when cyclical sectors or themes, for instance Technology, attract strong investor demand.

Figure 9: Performance of the Long/Short Portfolios based on the ETF Flow Signal



Source: Trackinsight, Bloomberg & authors' calculations.

The long–short strategy achieves a Sharpe ratio of 0.98, substantially exceeding the 0.14 Sharpe ratio of the equally weighted S&P 500 benchmark. This finding underscores the informational value of ETF flow–based signals for equity portfolio management.

Table 5: Performance Statistics of the Long–Short Strategy and Benchmarks

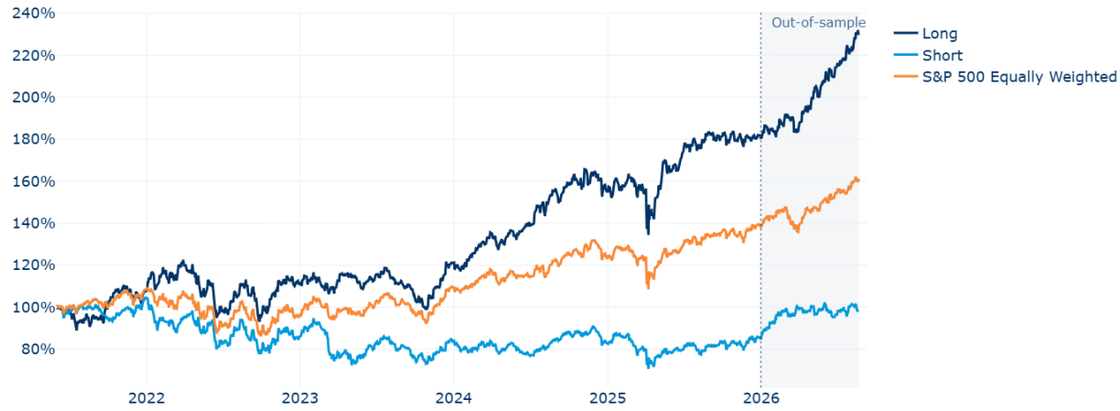
	Long-Short	S&P 500 Equally Weighted	Risk-Free Rate
Annualized Return (%)	17.57%	5.82%	3.42%
Annualized Volatility (%)	14.38%	17.35%	0.15%
Maximum Drawdown (%)	-15.96%	-21.59%	0.00%
Sharpe Ratio	0.98	0.14	0.00
Beta	0.06	1.00	0.00

3.3.2 Out-of-sample performance since model specification

The signal specification and portfolio construction rules described above were fixed using data available through December 2025. We therefore extend the analysis from January 2026 to August 2026 without re-estimating the model or modifying the signal definition, ranking methodology, portfolio size or rebalancing frequency. This post-sample period provides an additional assessment of the signal on data that were not used in its original specification.

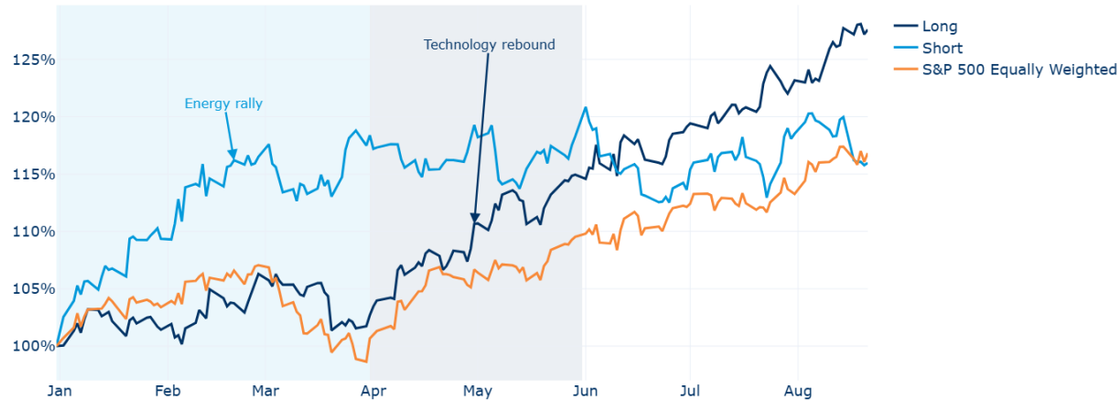
Figure 10 extends Figure 8 over the whole 2021–2026 window, the shaded area marking the out-of-sample period, and Figure 11 shows that period on its own.

Figure 10: Performance of the Long and Short Portfolios, In-Sample and Out-of-Sample



Source: Trackinsight, Bloomberg & authors' calculations.

Figure 11: Performance of the Long and Short Portfolios over the Out-of-Sample Period



Source: Trackinsight, Bloomberg & authors' calculations.

Table 6 shows that the long leg outperformed the equally weighted benchmark by 10.8 percentage points over the eight-month period, while the short leg performed broadly in line with the benchmark. Two sector exposures help explain the strategy's performance. In the opening months of 2026, the short portfolio was heavily exposed to energy stocks, following a period of strong ETF inflows into the sector in the aftermath of the U.S. intervention in Venezuela. The subsequent escalation of tensions in the Middle East, however, triggered a sharp increase in crude oil prices, sustaining both investor flows and energy-sector returns rather than allowing the expected reversal to materialize. This exogenous shock therefore temporarily disrupted the mean-reversion mechanism underlying the signal. From January to March, the short portfolio gained 18.4%, compared with only 0.7% for the equally weighted benchmark, resulting in a 13.7% loss for the long–short strategy.

The strategy subsequently recovered, supported by its changing exposure to technology stocks. The long portfolio was significantly exposed to the technology sector in April and May, before the signal reversed and turned negative on the sector in July, shortly before the market correction. From April onwards, the long portfolio returned 24.3%, compared with 16.1% for the equally weighted benchmark, while the long–short strategy gained 25.8%. This episode illustrates the ability of the signal to adjust to changing flow dynamics, shifting from a positive to a negative exposure as investor positioning evolved.

Table 6: Out-of-Sample Performance Statistics (January to August 2026)

	Long Leg	Short Leg	Long-Short	S&P 500 Equally Weighted
Cumulative Return (%)	27.59%	15.97%	8.54%	16.82%
Annualized Return (%)	46.78%	26.29%	13.77%	27.75%
Annualized Volatility (%)	12.95%	15.72%	18.80%	11.99%
Maximum Drawdown (%)	-4.63%	-6.90%	-13.45%	-7.86%
Sharpe Ratio	3.33	1.44	0.54	2.01
Beta	0.69	0.72	-0.03	1.00

3.3.3 Case studies

Having established the aggregate performance of the strategy, we now examine its behaviour at the thematic and sectoral level. The objective of this section is to assess whether the signal captures economically meaningful trends

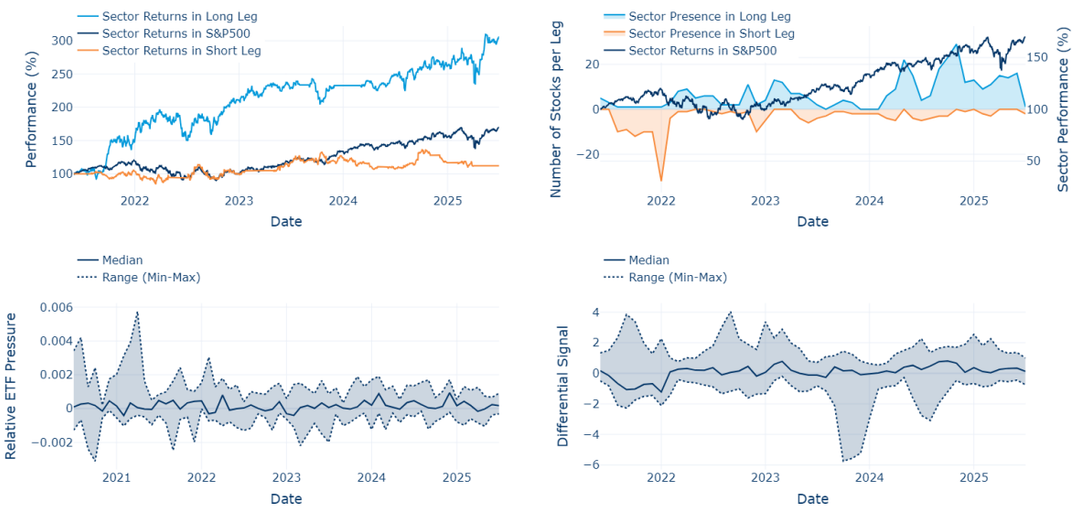
within major thematic segments of the equity market and whether portfolio exposures adjust in a manner consistent with the underlying return dynamics.

To link companies to sectors and themes, we assign each company to the sector/theme to which it is most exposed, based on Trackinsight's tagging at the ETF level.

We first consider the Artificial Intelligence & Big Data theme, comprising 73 stocks that were constituents of the S&P 500 at some point during the five-year sample period. This sector provides a particularly informative setting given its pronounced performance cycles and persistent ETF-related capital flows.

Figure 12 presents four complementary panels. The top-left panel displays the cumulative returns of the basket of Artificial Intelligence & Big Data constituents within the S&P 500, together with the average performance of the stocks belonging to the theme that are included in the long leg and, respectively, in the short leg, in order to assess the strategy's stock-selection ability. The top-right panel displays the number of components of the sector held in each leg alongside the cumulative performance of the sector within the S&P 500, to assess the strategy's timing capacity. The bottom-left panel displays the cross-sectional median and min–max range of ETF pressure across sector constituents. The bottom-right panel displays the corresponding cross-sectional median and min–max range for the standardized pressure differential signal S_m defined in Equation 9.

Figure 12: Artificial Intelligence & Big Data Sector: Performance, Exposures, and Flow-Based Measures



Source: Trackinsight, Bloomberg & authors' calculations.

The top-left panel indicates that the long leg captures the major upward phases of the sector and materially outperforms the equally weighted sector benchmark during prolonged upswings. Symmetrically, the short leg exhibits pronounced underperformance during sector expansions and tends to identify stocks that subsequently lag the broader thematic universe. This asymmetric behaviour suggests that the signal effectively differentiates between persistent inflow-driven strength and more transient pressure dynamics.

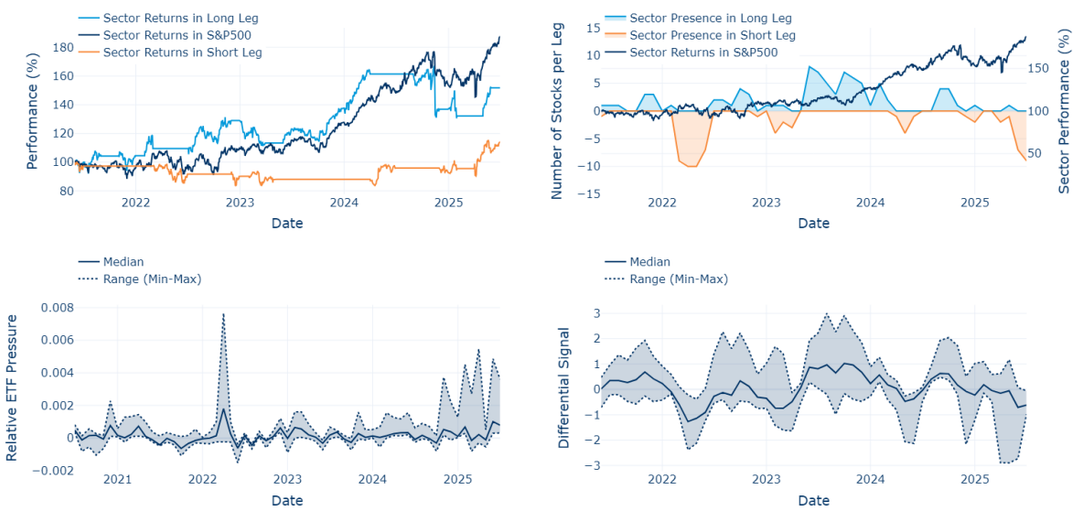
The top-right panel provides further information on portfolio construction. Periods of strong sector momentum are associated with a deliberate and economically coherent concentration of holdings. For example, at the end of 2022, 64% of the short portfolio (32 stocks) consisted of Artificial Intelligence & Big Data constituents. This elevated exposure preceded a marked relative decline in sector performance. By contrast, from mid-2024 through 2025, the long leg's exposure increased substantially, reaching 58% of the long portfolio (29 stocks) during a period of prolonged sector appreciation. These allocation patterns indicate that the strategy adjusts exposures consistently with evolving ETF pressure dynamics.

The bottom panels clarify the informational role of the underlying measures. The ETF pressure envelope reflects fluctuations in normalised flow intensity across sector constituents. Although informative regarding dispersion, this measure alone does not fully reveal directional structure. In contrast, the standardized pressure differential signal exhibits clearer trend formation. Periods in which short-term pressure dominates longer-term accumulation coincide with deteriorating relative performance and increasing short exposure, whereas phases characterised by strong twelve-month accumulation and subdued recent pressure align with expanding long allocations.

Taken together, these panels show the internal mechanics of the signal within a long-horizon investment framework. At each rebalancing date, the strategy incorporates the cumulative information embedded in one year of ETF activity while penalising short-term pressure imbalances. Portfolio decisions therefore emerge from the interaction of the two components: securities experiencing disproportionate short-term inflows relative to longer-term trends are de-emphasised or shorted, whereas those exhibiting sustained accumulation without excessive recent pressure are favoured.

To assess whether these dynamics extend to sectors with different structural characteristics, we next examine the Aerospace & Defence sector. This universe is smaller, comprising 13 stocks that were constituents of the S&P 500 at some point over the sample period. In recent years, the sector has experienced substantial inflows driven by heightened geopolitical tensions and increased defence spending. Its policy-sensitive and capital-intensive profile offers a distinct testing ground for the signal's responsiveness.

Figure 13: Aerospace & Defence Sector: Performance, Exposures, and Flow-Based Measures



Source: Trackinsight, Bloomberg & authors' calculations.

Immediately following the outbreak of the war in Ukraine, substantial inflows into the defence theme were observed. Given the unusually strong nature of these flows, companies exposed to the theme were added to the short leg. From late 2022, the Aerospace & Defence sector recorded a pronounced acceleration in ETF inflows, accompanied by strong cumulative performance through to the end of 2024. Over this period, the long leg closely mirrored the sector's upward trajectory, capturing most of its appreciation. This pattern is consistent with the signal identifying stocks exposed to sustained capital allocation within the thematic universe.

From early 2025, ETF flows became more volatile. This shift was reflected in increased variability of the standardised pressure differential signal and resulted in more frequent portfolio rebalancing. Sector exposure within the long leg was periodically reduced and, at times, transferred to the short leg, indicating the signal's sensitivity to reversals in short-term pressure relative to longer-term accumulation dynamics.

As Aerospace & Defence companies included in the short leg significantly underperformed the sector's average performance, the long leg began to underperform as flows became more erratic. The dispersion envelope indicates a widening in cross-sectional flow heterogeneity beginning in late 2022, followed by elevated variability from early 2025 onwards. The signal responds asymmetrically to these changes, strengthening when twelve-month accumulation prevails and adjusting rapidly when short-term pressure intensifies. The renewed increase observed towards the end of 2025 is consistent with a re-emergence of sustained accumulation.

Taken together, the two case studies reveal complementary features of the framework. In a broad, innovation-driven sector such as Artificial Intelligence & Big Data, the signal captures medium-term thematic cycles and reallocates capital progressively. In a narrower and policy-sensitive sector such as Aerospace & Defence, it reacts more visibly to sharp inflow accelerations and subsequent volatility.

Across both settings, portfolio exposures evolve coherently with the cross-sectional structure of ETF pressure. Nevertheless, these sectoral behaviours should be interpreted within a broader multi-factor investment environment in which macroeconomic conditions, valuation dynamics, and cross-sector relative opportunities jointly influence allocation decisions. The interaction between twelve-month accumulation and three-month imbalances appears central to identifying both persistent trends and potential inflection points, reinforcing the interpretation of ETF flows as a meaningful contributor to cross-sectional equity return dispersion.

3.3.4 Robustness Checks and Implementation Considerations

To assess the robustness of our signal, we lag the monthly rebalancing date by one to five business days, and recompute the backtest. This exercise verifies that the signal does not rely on a precisely timed execution date and remains stable under small implementation delays. The resulting variations remain limited. For the long leg, the maximum difference relative to the baseline specification amounts to 2.20% in annualised return, 0.12% in volatility, and 1.77% in maximum drawdown. For the short leg, the corresponding maximum differences are 0.87%, 0.21%, and 4.32%, respectively. These modest changes indicate that the signal remains informative even when rebalancing occurs with small operational delays.

The portfolio construction intentionally remains simple. Stocks are ranked independently at each rebalancing date using a cross-sectional indicator. Because the ranking is recomputed every month based on contemporaneous ETF flow fluctuations, the portfolio composition may change substantially from one date to the next. In addition, the strategy does not impose sector allocation constraints and uses equal weighting. These choices are deliberate and aim to isolate the informational content of the signal rather than to optimise portfolio construction.

To assess the resulting trading activity, we compute portfolio turnover, defined as:

$$Turnover_t = \frac{1}{2} \sum_{i=1}^{N} |w_{i,t} - w_{i,t-1}| \,. \tag{10}$$

where $w_{i,t}$ denotes the portfolio weight of stock i at rebalancing date t, and $w_{i,t-1}$ its weight the day before rebalancing.

Average monthly turnover amounts to 36.16% for the long leg and 42.20% for the short leg. These elevated levels reflect the cross-sectional nature of the signal and the size of the portfolio. While the exercise confirms the

informational relevance of ETF flow signals, it also indicates that practical implementations would require careful consideration of transaction costs and portfolio construction techniques aimed at reducing turnover.

3.4 Our signal in the Fama-French Framework

Because our strategy may resemble an ETF-flow-based momentum signal, there is a risk that it simply represents an alternative construction of the traditional momentum factor. We therefore evaluate our signal within the Fama–French asset pricing framework to determine whether it delivers incremental explanatory power beyond established risk factors.

The Fama–French five-factor model (Fama and French 2015) extends both the CAPM and the earlier three-factor model to better explain cross-sectional differences in stock returns. It posits that expected returns are driven by exposure to five systematic risk factors:

- MKT (Market factor) : the excess return of the market portfolio over the risk-free rate.
- SMB (Small Minus Big) : the size premium, capturing the historical outperformance of small-cap stocks relative to large-cap stocks.
- HML (High Minus Low) : the value premium, reflecting the tendency of high book-to-market ("value") firms to outperform low book-to-market ("growth") firms.
- RMW (Robust Minus Weak) : the profitability factor, indicating that firms with strong operating profitability tend to earn higher returns than less profitable firms.
- CMA (Conservative Minus Aggressive) : the investment factor, showing that firms that invest conservatively tend to outperform those that invest aggressively.

The model's core insight is that differences in average returns across stocks can largely be explained by their sensitivities (betas) to these five common risk factors, which proxy for fundamental dimensions of systematic risk related to size, valuation, profitability, and investment behaviour.

We augment our analysis by incorporating the momentum factor introduced by (Carhart 1997). Carhart proposes the WML (Winners Minus Losers) factor to capture return persistence documented in mutual fund performance. We include this factor alongside the Fama–French five-factor framework, as it is conceptually related to our ETF flow–based signal and allows us to assess whether our results simply reflect conventional momentum effects.

The analysis presented in this section relies on Fama–French five-factor model data retrieved from Kenneth French's data library [1].

We define ITF (Intertemporal Flow Factor) as the long-short portfolio constructed in 3.3. The portfolio goes long stocks that receive strong inflows from thematic and sectoral ETFs over a 12-month horizon but relatively weak inflows over the most recent three months. It goes short stocks exhibiting the opposite pattern : low 12-month inflows and high recent 3-month inflows.

The ITF portfolio displays low correlations with the Fama–French five factors, supporting the view that it represents an independent source of systematic return variation in the S&P 500. Notably, its correlation with the momentum factor (WML) remains modest, indicating that the strategy is not merely a repackaging of traditional momentum.

Table 7: Correlation Matrix

	ITF	Mkt-RF	SMB	HML	RMW	CMA	WML
ITF	1.00						
Mkt-RF	0.09	1.00					
SMB	0.05	0.18	1.00				
HML	0.08	-0.43	0.14	1.00			
RMW	-0.19	-0.26	-0.49	0.09	1.00		
CMA	0.13	-0.45	-0.05	0.78	0.20	1.00	
WML	0.20	-0.14	-0.22	0.26	0.04	0.34	1.00

To assess whether our ITF factor has explanatory power for S&P 500 returns, we next evaluate its contribution within a standard asset pricing framework by estimating time-series regressions for each stock in the index. Specifically, for each stock, we estimate regressions of excess returns on the Fama–French five factors and the momentum factor, augmented by the ITF factor.

Accordingly, we use a first regression model as a benchmark for return explanation :

$$r_i - R_f = \alpha_i + \beta_{MKT} \cdot (R_m - R_f) + \sum_{g \in \mathcal{F}} \beta_{i,g} \cdot R_g + \epsilon_i$$
$$\mathcal{F} = \{SMB, HML, RMW, CMA, WML\} \tag{11}$$

where r stands for the company's return, $\beta_{i,g}$ the sensitivity of company i to the factor g, R_g the return of the factor g, and R_f stands for the risk-free rate.

To assess the marginal contribution of our ITF factor, we define a second model as follows:

$$r_i - R_f = \alpha_i + \beta_{MKT} \cdot (R_m - R_f) + \sum_{g \in \mathcal{F}} \beta_{i,g} \cdot R_g + \beta_{ITF} \cdot R_{ITF} + \epsilon_i \tag{12}$$

We begin by examining the statistical significance of each factor across stocks. Table 8 reports the proportion of stocks for which each factor is significant at the 10%, 5%, and 1% levels.

[1] Data is available here.

Table 8: Share of Statistical Significance across Stocks (%)

Factor	10% level	5% level	1% level
ITF	36.32	27.71	14.80
Mkt-RF	98.45	98.28	97.76
SMB	69.54	63.34	51.81
HML	68.50	61.10	50.09
RMW	58.00	51.81	39.59
CMA	53.87	48.71	36.14
WML	72.63	67.99	61.62

Consistent with the academic literature, we find that traditional factors and the market factor are significant for a large fraction of stocks. The ITF factor is significant for more than one-third of the firms in our sample, suggesting that it possesses meaningful explanatory power.

To complement this analysis, we also assess the impact of including the ITF factor on the adjusted R^2. We compare the adjusted R^2 from Equation 12 and Equation 11 to quantify the incremental increase in the explained variance of returns. We use the adjusted R^2 rather than the R^2, as the latter mechanically increases with the inclusion of additional regressors.

Table 9: Incremental Explanatory Power of ITF (%)

	All	10% level	5% level	1% level
Usual Factors adjusted R^2	37.77	36.98	37.85	37.12
With ITF adjusted R^2	37.96	37.59	38.58	38.18
Improvement	0.19	0.61	0.73	1.06

The inclusion of ITF leads to a consistent increase in explanatory power. While the average improvement is modest across all stocks, it becomes more pronounced for stocks where ITF is statistically significant, reaching over 1% at the 1% level. This indicates that ITF captures meaningful variation in returns precisely where the signal is most relevant.

Overall, these results suggest that ITF provides incremental explanatory power beyond standard factors, while remaining concentrated on a subset of stocks more directly exposed to ETF-driven flows.

3.5 The case of the European market

ETF ownership data and the correlogram analysis indicate that the influence of the ETF market on the European equity market appears to be more limited than in the United States. To confirm this intuition, we replicated the empirical framework on European equities. Specifically, we apply the regression specification detailed in Equation 6 to the constituents of the STOXX 600 index, using ETF-induced pressure measures constructed from thematic and sectoral ETFs. As in the U.S. analysis, we estimate Equation 6 separately for each of the twelve accumulation horizons, avoiding the simultaneous inclusion of highly correlated ETF-pressure measures constructed over overlapping windows.

A key structural difference between the U.S. and European markets concerns the relative importance of ETFs in the ownership structure. Whereas ETFs hold approximately 16% of U.S. large-cap equities, their ownership share in European large-cap stocks is closer to 6%. This substantial difference may affect the strength of the statistical relationship between ETF-induced flows and subsequent stock returns. In addition, the diffusion of ETFs occurred later in Europe than in the United States, potentially limiting the persistence and informational content of ETF flows over the sample period. Taken together, these institutional differences suggest that weaker statistical associations may be expected in the European setting.

Table 10 reports the estimation results across cumulative flow windows ranging from one to twelve months.

The results reveal a clear short-term effect but no statistically robust long-term relationship. At short horizons, coefficients are strongly negative and statistically significant, particularly over two and three month windows with significances at the 1% and 5% levels. This provides clear evidence of a mean reversion effect consistent with temporary price pressure induced by ETF flows. The effect remains detectable, though weaker, up to horizons of around six to seven months.

In contrast, no meaningful long-term effect can be identified. Beyond eight months, estimated coefficients are small and statistically insignificant, with confidence intervals systematically spanning both sides of zero. Unlike the U.S. case, there is no contiguous range of horizons over which coefficients are positive and significant, preventing any reliable identification of a long-horizon allocation effect.

The explanatory power of the regressions is higher than in the U.S. specification, with R^2 values around 31%, reflecting the dominant role of market-wide factors. However, this improvement does not translate into stronger flow-based explanatory power, as ETF-induced pressure contributes only marginally to cross-sectional return variation.

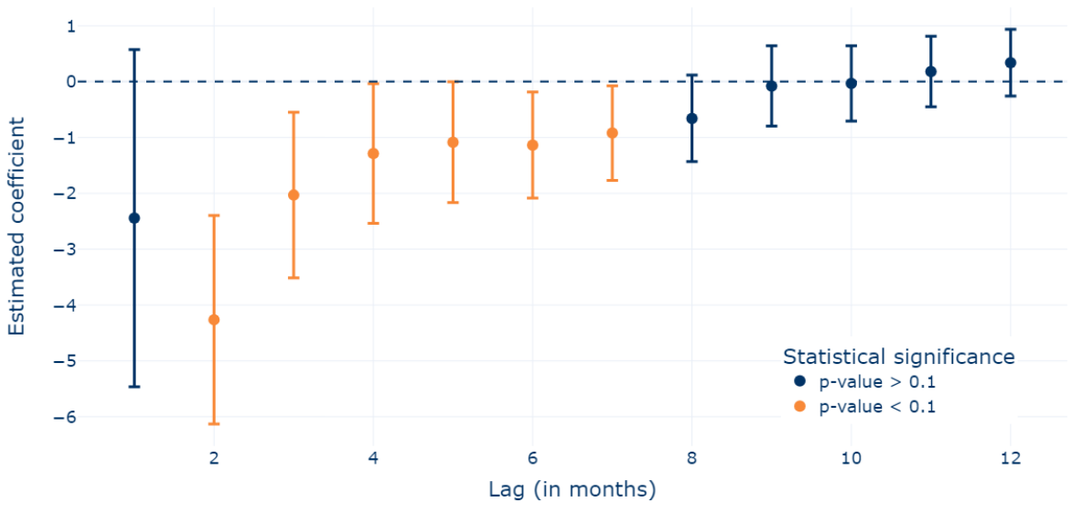
Figure 14 reports the estimated coefficients across horizons together with their 90% confidence intervals and indicates their statistical significance.

Table 10: Regression Results across Horizons (STOXX 600, Thematic and Sectoral ETFs)

Horizon	Coefficient	Standard Error	p-value	90% Confidence Interval	R^2	Observations
1	-2.44	1.84	0.18	[-5.47, 0.58]	0.31	24,344
2	-4.26***	1.13	< 0.01	[-6.13, -2.40]	0.31	24,339
3	-2.03**	0.90	0.02	[-3.51, -0.55]	0.31	24,333
4	-1.29*	0.76	0.09	[-2.54, -0.04]	0.31	24,327
5	-1.08*	0.66	0.10	[-2.17, 0.00]	0.31	24,322
6	-1.14*	0.58	0.05	[-2.09, -0.18]	0.31	24,317
7	-0.92*	0.52	0.07	[-1.77, -0.07]	0.31	24,314
8	-0.66	0.47	0.16	[-1.43, 0.12]	0.31	24,311
9	-0.08	0.44	0.86	[-0.80, 0.64]	0.31	24,307
10	-0.03	0.41	0.94	[-0.71, 0.64]	0.31	24,302
11	0.18	0.38	0.64	[-0.45, 0.81]	0.31	24,292
12	0.34	0.36	0.35	[-0.26, 0.94]	0.31	24,197

Notes: *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. All tests are two-sided.

Figure 14: Flow Coefficients and 90% Confidence Intervals by Horizon (STOXX 600, Thematic & Sectoral ETFs)



Source: Trackinsight, Bloomberg & authors' calculations.

The figure confirms this pattern. Short-horizon coefficients are negative and statistically significant, while longer-horizon estimates cluster around zero with wide confidence intervals, indicating the absence of a stable long-term effect.

Overall, the European evidence highlights a clear asymmetry relative to the U.S. market. While short-term price pressure effects are detectable, no statistically robust long-horizon relationship emerges. This suggests that ETF-induced flows remain too limited in scale to generate persistent cross-sectional return patterns. The results therefore reinforce the interpretation that the stronger effects observed in the U.S. are largely driven by the greater penetration of ETFs in the ownership structure of equities.

4 Discussion

While the empirical results highlight a clear relation between ETF flows and subsequent equity returns, several limitations should be acknowledged.

First, although our dataset covers a large number of ETFs, it does not include the entirety of global ETF assets under management. Measurement error in ETF ownership and in the construction of stock-level pressure indicators may therefore attenuate the documented relationships. In addition, the identification of thematic and sectoral ETFs relies on the classification framework provided by Trackinsight. Alternative classification schemes could reassign certain products across categories and potentially affect the magnitude and the horizons of the estimated effects.

Second, the empirical specification relies on fixed horizons to capture short- and longer-term flow dynamics. In our portfolio application, we use twelve-month and three-month windows for simplicity and transparency. However, other horizon choices may also be informative. Future work could explore more flexible specifications, including weighted combinations of horizons guided by the regression results.

Our portfolio implementation is intentionally simple. Portfolios are equally weighted and constructed purely from the cross-sectional ranking of the signal in order to isolate the informational content of ETF-induced pressure. In practical investment settings, alternative constructions may be preferable. Portfolio weights could incorporate market

capitalisation, liquidity characteristics, or signal intensity, while additional constraints such as sector allocation limits or turnover buffers could be introduced to reduce trading costs.

Another natural extension concerns the role of mutual funds. Although ETF flows into physically replicated ETFs are transparent and mechanically transmitted to underlying markets, mutual funds still represent a large share of delegated equity ownership. Comparing ETF and mutual fund induced pressures would help determine whether the documented horizon-dependent structure reflects a feature specific to ETF mechanics or a broader property of flow-driven markets.

Finally, our results suggest that the strength of ETF flow effects may depend on market structure. The clearer evidence observed in U.S. equities is consistent with the higher level of ETF penetration and the greater integration of ETFs into equity trading in that market. Examining how ETF ownership affects cross-sectional price dynamics across markets remains a possible direction for future research.

Another avenue for future research would be to develop a time-series ETF flow–based signal, rather than a cross-sectional one, thereby enabling the identification of unusual flows at the individual firm level. Practitioners may also be interested in ETF-based signals. In this context, statistical analysis and signal construction could be extended to thematic and sectoral ETFs to design trend-following strategies for ETF investors.

5 Conclusion

This paper studies the relation between ETF flows and underlying equity returns within the S&P 500 universe. Our empirical analysis shows that ETF-induced flows contain information about subsequent stock returns, with effects that vary across investment horizons. Short-term inflows tend to be followed by partial price reversals, whereas more sustained inflows from thematic and sectoral ETFs are associated with persistent return effects. These horizon-dependent dynamics are further supported by a sinusoidal specification linking past flows to subsequent returns. The short-term reversal effect is consistent with prior studies, although it operates over a slightly longer horizon in our setting, while the longer-term pattern suggests that ETF flows tend to reinforce prevailing market trends.

Our findings are particularly pronounced for thematic and sectoral ETFs. Flows originating from these products appear more informative in highlighting the relationship between flows and returns, likely reflecting tactical investor positioning and concentrated capital reallocation across segments of the equity market.

From a portfolio management perspective, ETF flow data therefore provide a useful basis for comparing stocks. In particular, the signal helps identify securities with particularly high exposure to ETF-induced pressure alongside their fundamental drivers, and whether this flow-induced pressure is currently supporting or weighing on their performance. Signals derived from ETF-induced pressure may therefore help equity investors identify stocks supported by persistent capital inflows, detect securities exposed to short-term flow-driven reversals, and distinguish between positive and negative flow effects. Our backtests indicate that the long portfolio outperforms the S&P 500 Equally Weighted benchmark, while the short portfolio materially underperforms, suggesting that the signal may be valuable both for stock selection and for identifying stocks to avoid.

Finally, a comparison between the ETF flow–based factor and traditional risk premia indicates limited correlation with the Fama–French factors and with momentum (WML). This suggests that ETF flow dynamics capture a distinct dimension of equity market behaviour that may complement existing factor-based allocation frameworks and provide economically meaningful information for practitioners. Augmenting the Fama–French model by incorporating our ITF factor shows that the ITF factor is significant for 36% of S&P 500 firms at the 10% level. Moreover, the adjusted R^2 increases by more than one percentage point for firms where the factor is highly significant. These results indicate that our factor enhances the explanatory power of return variation.

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7 Declaration of interest

The authors are employed by Kepler Cheuvreux, which is the majority shareholder of Trackinsight. Kepler Cheuvreux may develop investment products related to the subject matter of this research. The authors declare that this employment relationship did not influence the research design, analysis, interpretation of results, or conclusions.

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